



Deep Mean-Reversion: A Physics-Informed Contrastive Approach to Pairs Trading

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Abstract

Traditional pairs trading strategies often fail to identify stable mean-reverting relationships amid the complex dynamics characterizing modern financial markets, resulting in unstable spreads and reduced efficacy of conventional, rule-based execution methods. To overcome these intrinsic limitations, this study introduces **ORCA** (Ornstein-Uhlenbeck Reversion and Contrastive Arbitrage), a novel framework that seamlessly integrates deep representation learning and a rigorous financial dynamics model into a unified training paradigm. Central to ORCA is a physics-informed regularization approach designed to identify asset clusters that exhibit not merely similarity, but intrinsic *tradability* characterized by robust dynamic properties. ORCA concurrently optimizes a contrastive learning module alongside a Physics-Informed Neural Network (PINN) module, where the latter serves as a regularizer enforcing cluster formation consistent with the statistical dynamics of a stable Ornstein-Uhlenbeck process. Consequently, ORCA systematically produces asset clusters with inherently superior mean-reversion characteristics. Empirical analysis conducted on the NYSE dataset demonstrates the practical effectiveness of ORCA: applying a simple mean-reversion trading strategy with a static threshold to ORCA-generated clusters significantly outperforms strategies employing clusters derived via alternative benchmark methodologies. These findings position ORCA as a new benchmark methodology in the realm of structure-aware statistical arbitrage. A comprehensive overview of empirical results is provided in Figure 1.

CCS Concepts

• **Computing methodologies** → **Learning latent representations**; • **Applied computing** → **Economics**; *Business intelligence*.

Keywords

Contrastive Learning, Physics-Informed Neural Networks, Mean-Reversion, Pairs-trading, Portfolio

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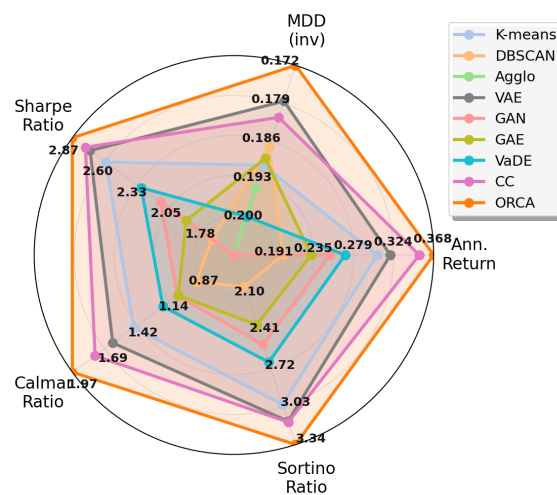


Figure 1: Overall performance comparison.

1 Introduction

Pairs trading [5] is a market-neutral strategy that leverages the mean-reverting nature of spreads between assets sharing common economic links. Its principle involves identifying asset pairs whose prices have historically co-moved and executing arbitrage trades—taking a long position in the underperforming asset and a short position in the outperforming one—when their price spread diverges from statistical equilibrium.

However, the success of this traditional approach faces two fundamental challenges. First, the pair selection process often fails to capture complex, nonlinear relationships between assets. Traditional statistical methods frequently rely on superficial similarities in raw feature space, which can lead to the selection of unstable pairs in dynamic market environments [29]. Second, the low quality of the selected pairs inherently amplifies instability during the trade execution phase. Spreads derived from these suboptimal pairs are often noisy and exhibit weak mean-reverting characteristics. Consequently, when a simple rule-based method is applied to these unstable signals, it inevitably leads to inefficient outcomes, such as being whipsawed by noise or failing to capture faint signals.

To transcend these limitations, we introduce **ORCA** (Ornstein-Uhlenbeck Reversion and Contrastive Arbitrage), a novel framework that synergistically integrates deep representation learning with an economic theory-based financial model within a single, unified training process. ORCA's core innovation lies in identifying

asset clusters that are not merely similar, but are inherently *tradable* with strong dynamic properties, a concept we enforce through physics-informed regularization.

Our framework simultaneously optimizes a contrastive learning module and a Physics-Informed Neural Network (PINN) module [25]. While contrastive learning [2] elucidates the latent structures within the data to build robust asset representations, the PINN module acts as a regularizer. It compels the clustering framework to form groups whose members collectively adhere to the statistical properties of a stable Ornstein-Uhlenbeck (OU) stochastic differential equation. Through this integrated learning process, ORCA evolves to produce higher-quality asset clusters with intrinsically enhanced mean-reverting characteristics.

The main contributions of this paper are as follows:

- A novel end-to-end framework integrating contrastive learning with Physics-Informed Neural Networks (PINN) is proposed to systematically identify intrinsically tradable asset clusters.
- A new learning paradigm leveraging PINNs as a regularization mechanism is introduced, compelling the clustering process to adhere strictly to established financial engineering dynamics, specifically the Ornstein-Uhlenbeck process.
- Comprehensive empirical validation using extensive real-world NYSE market data demonstrates that conventional mean-reversion trading strategies significantly outperform alternative benchmark methodologies when applied to clusters identified by the proposed ORCA framework.

2 Related Works

2.1 Contrastive Learning

Contrastive learning [24] is a promising paradigm in self-supervised learning, recently achieving new benchmarks in representation learning in fields such as computer vision [2, 10, 15] and natural language processing [6]. The fundamental idea of this approach is to map the original data into a representation space, wherein the similarities of positive pairs are maximized while those of negative pairs are minimized [12]. We primarily follow the SimCLR's [2] temperature-scaled cross-entropy loss (NT-Xent). Building a positive pair through data augmentation is essential for contrastive learning. In computer vision, positive pairs are created by performing random transformations on the same image, such as cropping, flipping, and jittering [10, 15]. Natural language processing uses methods such as dropout [6] and discrete masking [3] to build positive pairs. Unlike the above two areas, we take a different approach to building positive pairs for tabular data by using random masking, which randomly erases values as zero, and Gaussian noise.

2.2 Physics-Informed Neural Networks in Finance

Physics-Informed Neural Networks (PINNs) have emerged as a powerful tool for modeling systems governed by differential equations [25]. The core idea of the PINNs paradigm is to incorporate the residual of a differential equation into the neural network's loss function, thereby compelling the model to not only fit the data but also adhere to the underlying physical law. This paradigm has been

successfully applied in financial engineering, particularly for option pricing. For instance, PINNs have been utilized to solve the Black-Scholes Partial Differential Equation (PDE) in data-scarce regions, price exotic options, and calibrate market volatility surfaces [7, 13]. These studies demonstrate the effectiveness of PINNs in modeling the complex dynamics of financial markets. Our work applies this idea to pairs trading by incorporating the Ornstein-Uhlenbeck (OU) Stochastic Differential Equation (SDE), which governs the mean-reverting dynamics of a spread, into the learning process.

2.3 Machine Learning in Pairs Trading

Recent developments in pairs trading have focused on integrating machine learning techniques to enhance performance. Unsupervised learning methods, such as clustering algorithms, have been employed to identify complex, non-linear relationships that traditional statistical measures may overlook [26]. For example, K-means clustering has been utilized to group assets based on similarities in their feature representations, providing a more nuanced approach than linear models. Meanwhile, reinforcement learning (RL) has also been explored as a promising avenue in pairs trading [1, 18, 31]. RL agents optimize trading strategies by learning directly from market dynamics via a reward mechanism. However, RL can often overfit to specific asset pairs and its need for direct interaction with the environment can make it challenging to apply to a large universe of assets.

Most existing approaches tend to focus on either improving pair selection or enhancing trading execution. In this paper, we propose to unify robust, data-driven pairing via contrastive learning with financial dynamic modeling through the integration of a PINN framework.

3 Method

Our proposed framework, ORCA, is designed to identify inherently tradable asset clusters through a unified, end-to-end training process. This is achieved through three interconnected components: (1) a Backbone Architecture that extracts rich, dynamic representations for each asset from raw data; (2) a Physics-Informed Contrastive Clustering module that uses a novel unified loss function to simultaneously learn asset similarities and enforce that the resulting clusters adhere to the principles of mean-reversion, specifically the Ornstein-Uhlenbeck (OU) process; and (3) a Cluster-Based Trading Strategy that leverages these high-quality clusters to execute trades. The overall architecture is depicted in Figure 2.

3.1 Backbone Architecture

The backbone is designed to extract rich asset representations using a feature encoder and a bidirectional transformer. Since deep learning models can underperform boosting methods on tabular data [11, 27], we employ a specialized feature encoder based on Piece-wise Linear Encoding (PLE), a modern binning approach with proven success in tabular deep learning [9, 21]. Unlike continuous embeddings such as linear layers, PLE explicitly preserves ordinal relationships and creates informative representations robust to outliers. The resulting tokens are processed by a bidirectional transformer [28], which excels at capturing the complex relationships between all input features, such as firm characteristics and

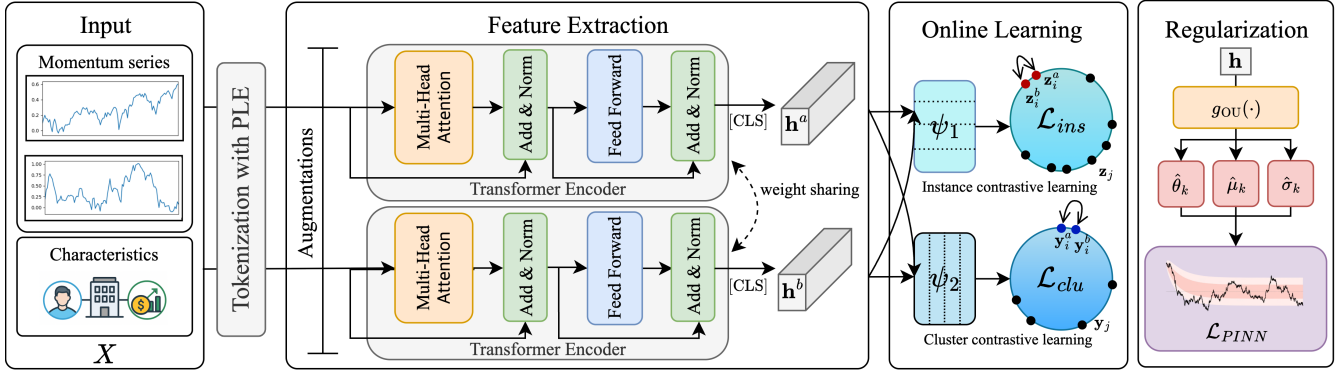


Figure 2: Overall architecture of ORCA. Input features X are tokenized via PLE and augmented into two views, which are processed by a shared transformer encoder to produce representations h^a and h^b . The model optimizes three objectives: instance-level contrast $\mathcal{L}_{ins}$, cluster-level contrast $\mathcal{L}_{clu}$, and physics-informed regularization $\mathcal{L}_{PINN}$ that enforces Ornstein-Uhlenbeck dynamics with parameters $(\hat{\theta}_k, \hat{\mu}_k, \hat{\sigma}_k)$.

price momentum [17, 30]. This backbone ultimately outputs a representation vector h for each asset.

3.1.1 Tabular feature encoder. The feature encoder transforms structured tabular data, where the k -th numerical feature x_k is encoded as $\tilde{x}_k = \text{enc}_k(x_k) \in \mathbb{R}^d$. Here, $\text{enc}_k(\cdot)$ is the encoder for the k -th feature and d is the embedding dimensionality. The encoding process begins with binning. First, we normalize all features using a standard scaler. Then, for each feature k , we define T bins, $B_1, \dots, B_T$, with boundaries $[b_0, b_1), \dots, [b_{T-1}, b_T]$. The first bin $B_1 = (-\infty, b_1)$ and the last bin $B_T = [b_{T-1}, \infty)$ are unbounded to ensure complete coverage of the feature domain. For a given scalar value x , the PLE function transforms it into a T -dimensional vector $\text{PLE}(x) = [e_1, \dots, e_T]$. The value of each component e_t is determined by the following conditions:

$$e_t = \begin{cases} 0, & \text{if } x < b_{t-1} \text{ and } t > 1 \\ 1, & \text{if } x \geq b_t \text{ and } t < T \\ \frac{x - b_{t-1}}{b_t - b_{t-1}}, & \text{otherwise} \end{cases} \quad (1)$$

This encoding produces a cumulative representation: bins entirely below x_k are set to 1, the bin containing x_k receives a fractional value based on its relative position within that bin, and bins above x_k are set to 0. This creates a dense, ordered representation of the numerical feature. Finally, this T -dimensional vector is passed through a trainable linear layer to produce the final d -dimensional embedding that is fed into the transformer:

$$\tilde{x}_k = \text{Linear}(\text{PLE}(x_k)) \quad (2)$$

3.1.2 Feature transformer. The feature transformer compresses the characteristics of each asset into a single vector. To improve the modeling of the complex relationships between asset characteristics, we use a bidirectional transformer. This transformer is internally structured with a multi-headed self-attention mechanism, which allows it to reflect multifaceted information, including the relationship between firm characteristics and price momentum, from different perspectives within the feature vector. As a result, the feature vector is embedded in the feature vector space with

more comprehensive attributes. The mathematical formulation of the multi-headed attention mechanism in our model is as follows:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (3)$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

Similar to BERT [4], the successful transformer architecture in natural language processing, we append a special [CLS] token with a learned embedding to the beginning of each data sample. Let $\mathbf{x}_i = [\text{[CLS]}, \tilde{x}_1^i, \tilde{x}_2^i, \dots, \tilde{x}_n^i]$ be the input sequence for a single asset, composed of its features $\tilde{x}_k^i$ encoded by binning. Let $f(\cdot)$ be the feature transformer. The input sequence $\mathbf{x}_i$ is transformed into a sequence of hidden states, and the final representation for the asset, h_i , is taken from the output corresponding to the [CLS] token's position via $h_i = f(\mathbf{x}_i)[\text{CLS}]$. This [CLS] token's output vector summarizes the information from all feature tokens through the self-attention mechanism.

3.2 Physics-Informed Contrastive Clustering

Using the representation vector h from the backbone, we train our model by optimizing a unified loss function with three objectives: (1) learning instance-level representations, (2) forming a cluster-level structure, and (3) ensuring the financial-dynamic validity of the formed clusters.

3.2.1 Contrastive Representation Learning. Our representation learning is based on contrastive learning performed at two simultaneous levels: the **instance** level and the **cluster** level. This dual approach guides the model to learn the unique characteristics of individual assets while concurrently forming a coherent macroscopic structure among groups of assets.

First, instance-level contrastive learning trains the model to learn a unique representation for each asset. The success of this hinges on the construction of positive pairs through effective data augmentation. To preserve the inherent characteristics of tabular

data while obtaining superior representation vectors, we employ two relatively simple augmentation techniques: random masking (ϕ_{mask}), which sets a fraction of feature values to zero, and the addition of Gaussian noise (ϕ_{Gauss}). These augmentations mirror real-world data imperfections in financial markets, such as missing fundamental data and noisy price observations, ensuring robustness without introducing additional complexity.

An original input $\mathbf{x}$ is transformed into two different views, $\mathbf{x}^a = \phi_{\text{mask}}(\mathbf{x})$ and $\mathbf{x}^b = \phi_{\text{Gauss}}(\mathbf{x})$. These augmented inputs are passed through the backbone architecture $f(\cdot)$ and a non-linear projection head $g_{\text{ins}}(\cdot)$ to be mapped into final representation vectors $\mathbf{z}^a$ and $\mathbf{z}^b$. The objective of instance contrastive learning is to maximize the similarity between $\mathbf{z}^a$ and $\mathbf{z}^b$, which originate from the same source, while minimizing their similarity to all other samples in the minibatch. Following SimCLR, the loss function for a given sample $\mathbf{x}_i^a$ is defined as:

$$\ell_i^a = -\log \frac{\exp(s(\mathbf{z}_i^a, \mathbf{z}_i^b)/\tau)}{\sum_{j=1}^N [\exp(s(\mathbf{z}_i^a, \mathbf{z}_j^a)/\tau) + \exp(s(\mathbf{z}_i^a, \mathbf{z}_j^b)/\tau)]}, \quad (4)$$

where $\mathbf{z} = g_{\text{ins}}(f(\mathbf{x}))$, $s(\cdot, \cdot)$ is the cosine similarity, and τ is a temperature hyperparameter. The total instance loss, $\mathcal{L}_{\text{ins}}$, is computed as the average over all positive pairs:

$$\mathcal{L}_{\text{ins}} = \frac{1}{2N} \sum_{i=1}^N (\ell_i^a + \ell_i^b). \quad (5)$$

However, instance-level contrastive learning alone may not be sufficient to form well-separated clusters. To complement this, we introduce cluster-level contrastive learning. The hidden state $\mathbf{h}$ from the backbone is fed into a separate cluster projection head, $g_{\text{clu}}(\cdot)$, which outputs a soft label vector $\mathbf{y} \in \mathbb{R}^M$, representing the probability of an asset belonging to each of the M clusters.

Cluster-level contrastive learning treats the cluster assignment probability vectors from the two augmented views of the same asset, $\mathbf{y}_i^a$ and $\mathbf{y}_i^b$, as a positive pair. The loss function is designed to maximize their similarity while minimizing their similarity to the cluster assignments of all other samples.

$$\hat{\ell}_i^a = -\log \frac{\exp(s(\mathbf{y}_i^a, \mathbf{y}_i^b)/\tau_c)}{\sum_{j=1}^N [\exp(s(\mathbf{y}_i^a, \mathbf{y}_j^a)/\tau_c) + \exp(s(\mathbf{y}_i^a, \mathbf{y}_j^b)/\tau_c)]}, \quad (6)$$

where τ_c is the cluster-level temperature parameter. To prevent a trivial solution where all assets collapse into a single cluster, we add a term that maximizes the entropy of the cluster assignments, $\mathcal{H}(Y)$. The final cluster loss is:

$$\mathcal{L}_{\text{clu}} = \frac{1}{2N} \sum_{i=1}^N (\hat{\ell}_i^a + \hat{\ell}_i^b) - \mathcal{H}(Y). \quad (7)$$

This dual contrastive learning process enables the model to form semantically consistent clusters based on feature similarity. However, clusters formed this way do not guarantee statistically significant trading opportunities, as they are based on feature similarity alone, not on the strength of their mean-reverting properties. To overcome this limitation, we introduce the physics-informed regularization, as described in the next section.

3.2.2 Physics-Informed Regularization. Clusters formed by feature similarity alone do not guarantee statistically significant trading opportunities. To bridge this gap, we introduce a physics-informed regularizer based on the Ornstein-Uhlenbeck (OU) process, which models mean-reverting dynamics. We assume that within each cluster k , asset returns follow a common OU process:

$$dr_t = \theta_k(\mu_k - r_t)dt + \sigma_k dW_t, \quad (8)$$

where r_t represents the return process, θ_k is the speed of mean reversion, μ_k is the long-term equilibrium return, and σ_k is the volatility for cluster k . Rather than assuming these parameters are static, our model dynamically predicts them for each cluster. This is achieved by first defining a cluster representation vector, $\bar{\mathbf{h}}_k$, which is the weighted average of the representations of assets assigned to that cluster:

$$\bar{\mathbf{h}}_k = \frac{\sum_i P_{ik} \cdot \mathbf{h}_i}{\sum_i P_{ik} + \epsilon}, \quad (9)$$

where P_{ik} is the soft probability that asset i belongs to cluster k . This aggregated representation is then fed into a dedicated parameter-generating neural network, $g_{\text{OU}}(\cdot)$, to yield the cluster's specific OU parameters:

$$[\hat{\theta}_k, \hat{\mu}_k, \hat{\sigma}_k] = g_{\text{OU}}(\bar{\mathbf{h}}_k). \quad (10)$$

This dynamic approach allows the model to derive a cluster's financial dynamics from the characteristics of its constituent members. The physics-informed regularization $\mathcal{L}_{\text{PINN}}$ measures how well these parameters explain the observed return dynamics of assets assigned to each cluster. For asset i with return series $r_i(t)$, the discretized residuals, $R_{ik}(t)$, which represents the deviation from the predicted OU dynamics of cluster k :

$$R_{ik}(t) = (r_i(t) - r_i(t - \Delta t)) - \hat{\theta}_k(\hat{\mu}_k - r_i(t - \Delta t))\Delta t \quad (11)$$

Assuming these residuals are Gaussian, the loss is formulated as the weighted negative log-likelihood. The final physics-informed regularization encourages the model to form clusters whose members exhibit coherent, predictable mean-reverting behavior:

$$\mathcal{L}_{\text{PINN}} = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K P_{ik} \left(\log(\hat{\sigma}_k) + \frac{\mathbb{E}_t[R_{ik}(t)^2]}{2\hat{\sigma}_k^2 \Delta t} \right), \quad (12)$$

where $\mathbb{E}_t[\cdot]$ denotes the expectation over time. This loss encourages the model to form clusters whose members exhibit coherent mean-reverting dynamics, effectively integrating financial principles into the clustering objective and producing groups that are inherently suitable for pairs trading strategies.

3.2.3 Unified Training Objective. The final model is trained end-to-end by minimizing a single, combined loss function that harmonizes representation learning, clustering, and financial dynamics:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ins}} + \alpha \mathcal{L}_{\text{clu}} + \beta \mathcal{L}_{\text{PINN}}, \quad (13)$$

where α and β are hyperparameters that balance the objectives. This unified training process guides the model to discover clusters that are not merely similar based on static features, but are inherently *tradable* with strong, verifiable mean-reverting characteristics.

3.3 Cluster-Based Trading Strategy

The core contribution of the ORCA framework is not to innovate the trading rules, but to fundamentally improve the quality of the asset groups being traded. Therefore, to demonstrate the superiority of the tradable clusters generated by ORCA, we deliberately apply a simple and conventional mean-reversion trading strategy. This strategy constructs spreads based on the prior month's return (mom_1) within each cluster and uses a static threshold for trade execution. The process is as follows. At each rebalancing period t , ORCA groups all assets into K clusters, $\{C_1, \dots, C_K\}$. For each cluster C_k , we determine long and short positions. First, we sort the assets within the cluster by their prior one-month return, R_{t-1} , to identify the top-performing (overvalued candidates) and bottom-performing (undervalued candidates) groups. The return differential between these two groups defines a momentum spread. A trading position is then initiated only when an asset's momentum spread exceeds a statistically significant level (e.g., a multiple γ of the standard deviation of all spreads). Long positions are taken in the assets that have underperformed the most relative to their cluster peers, and short positions are taken in the assets that have overperformed the most. By evaluating the performance of this simple, rule-based strategy when applied to ORCA's clusters versus its performance on clusters formed by other methodologies, we can clearly assess the practical value added by our physics-informed regularization. The detailed procedure of this strategy is summarized in Algorithm 1.

Algorithm 1 Cluster-Based Mean-Reversion Strategy

Require: Feature data $\mathbf{X}_t$, Return data R , Trained Model $f(\cdot)$, Threshold factor γ

Ensure: Period log return r_t

```

1: Cluster: Use model  $f(\mathbf{X}_t)$  to assign all assets to clusters  $\{C_1, \dots, C_K\}$ .
2: Momentum Spread: For each asset  $x$  in a cluster  $C_k$ , compute its momentum spread  $\Delta(x)$  based on its rank of  $R_{t-1}(x)$  within the cluster (see Section 3.3).
3:  $\sigma_\Delta \leftarrow \text{std}(\{\Delta(x) \mid \forall x\})$   $\triangleright$  Standard deviation of all spreads
4: Initialize asset signals  $s(x) \leftarrow 0$  for all  $x$ .
5: for each asset  $x \in \{C_1, \dots, C_K\}$  do
6:   if  $\Delta(x) < -\gamma \cdot \sigma_\Delta$  then  $\triangleright$  Long position
7:      $s(x) \leftarrow +1$ 
8:   else if  $\Delta(x) > \gamma \cdot \sigma_\Delta$  then  $\triangleright$  Short position
9:      $s(x) \leftarrow -1$ 
10:  end if
11: end for
12:  $\mathcal{A}_t \leftarrow \{x : s(x) \neq 0\}$ 
13: if  $|\mathcal{A}_t| > 0$  then
14:    $r_t \leftarrow \frac{1}{|\mathcal{A}_t|} \sum_{x \in \mathcal{A}_t} \log(1 + s(x) \cdot R_{t+1}(x))$ 
15: else
16:    $r_t \leftarrow 0$ 
17: end if
18: return  $r_t$ 

```

4 Experiments

In this section, we empirically validate the efficacy of the proposed method through a series of experiments.

4.1 Experimental Setup

4.1.1 Dataset. We obtain data from the CRSP/Compustat Merged (CCM) database via WRDS for the period 1999–2023. Our sample comprises all common stocks (CRSP share codes 10 and 11) listed on NYSE, AMEX, and NASDAQ exchanges. We compute monthly total returns from daily price and volume data, which we then transform into momentum features following Han et al. [14]. Specifically, we construct 24 momentum features for each stock-month observation:

$$\begin{aligned} \text{mom}_1 &= r_{t-1}, \\ \text{mom}_i &= \prod_{j=t-i}^{t-2} (1 + r_j) - 1, \quad \text{for } i \in \{2, \dots, 24\}, \end{aligned}$$

where r_j denotes the total return in month j . These features capture return dynamics across multiple time horizons. From Compustat, we incorporate key accounting variables updated quarterly: total assets (ATQ), total liabilities (LTQ), short-term debt (DLCQ), long-term debt (DLTTQ), shareholders' equity (SEQQ), cash and equivalents (CHEQ), net sales (SALEQ), net income (NIQ), operating income (OIADPQ), pretax income (PIQ), depreciation (DPQ), and diluted earnings per share (EPSPXQ). These fundamental indicators provide additional information beyond price momentum.

4.1.2 Backtesting. We conduct a monthly-rebalanced backtesting spanning 288 months from December 1999 to December 2023. At the end of each month, we follow a systematic procedure: (1) We group the entire asset universe into K clusters using ORCA and each of the baseline models. (2) Within each cluster, we compute a momentum spread for each asset based on its rank of the prior month's return (mom_1). (3) We take a long position in assets whose spread falls below $-\gamma \cdot \sigma$ and a short position in those whose spread exceeds $\gamma \cdot \sigma$, where γ is the threshold parameter and σ is the standard deviation of all spreads. For all experiments, we set $\gamma = 1.0$ and $K = 30$. All positions in the resulting portfolio are equally weighted, and a 10% stop-loss is applied to each individual position.

4.1.3 Baseline Models. To comprehensively evaluate the clustering performance of ORCA, we compare its performance against a diverse set of established methodologies. The baseline models are categorized into three groups:

- **Traditional Clustering:** Representative unsupervised algorithms based on the geometric distribution of asset features: **K-Means**, **DBSCAN**, and **Agglomerative Clustering (Agglo)**.
- **Generative Model-based Clustering:** Two-stage approaches that first learn a latent representation of the data, upon which a clustering algorithm is subsequently applied. We use a Variational Autoencoder (VAE) [19], a Generative Adversarial Network (GAN) [8], and a Graph Autoencoder (GAE) [20] to extract asset embeddings, and then apply K-Means to these embeddings to form the final clusters. This category also includes Variational Deep Embedding (VaDE) [16], which performs latent representation learning and clustering in an end-to-end fashion.

- **Contrastive-based Clustering:** Finally, we include a Contrastive Clustering (CC) [22] methodology, which represents a state-of-the-art data-driven approach, as a benchmark.

4.1.4 Evaluation Metrics. The performance of each strategy is assessed from multiple perspectives, focusing on risk-adjusted returns. We measure the portfolio performance using a suite of standard financial metrics: Annualized Return (**AR**), Annualized Volatility (**Vol**), Maximum Drawdown (**MDD**), and key risk-adjusted measures including the Sharpe Ratio (**Sharpe**), Calmar Ratio (**Calmar**), and Sortino Ratio (**Sortino**).

4.1.5 Implementation. All input features are normalized using a StandardScaler prior to training. Our model tokenizes the 36 normalized features into 64 bins, which are then embedded into a 128-dimensional hidden space. The transformer encoder consists of 2 layers with 8 attention heads. We train the model using the AdamW [23] optimizer with a learning rate of 0.002 and a batch size of 1024. For the unified loss function, $\mathcal{L}_{\text{total}}$, the weights for the $\mathcal{L}_{\text{clu}}$, α , and the $\mathcal{L}_{\text{PINN}}$, β , are both set to 1.0. For data augmentation, we use a standard deviation of 0.1 for the Gaussian noise and a 10% masking ratio. All experiments were conducted on a single NVIDIA RTX 3090 GPU with a fixed random seed of 42 for reproducibility. The backtest follows a monthly rebalancing schedule, where positions are entered at the beginning of each month and liquidated at the end.

4.2 Main Results

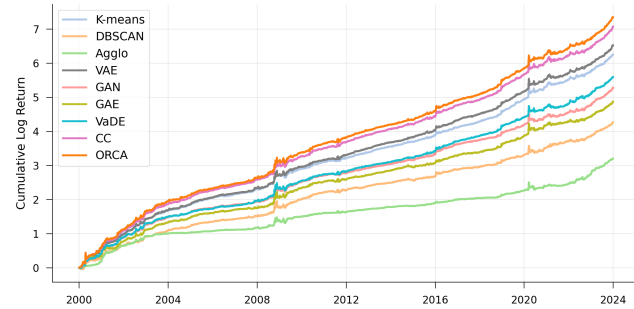


Figure 3: Cumulative returns of trading strategies from 2000 to 2023.

As shown in Table 1, the proposed model, ORCA, achieves superior performance over benchmark clustering methods across major metrics, with the exception of volatility. This demonstrates the superior ability of our framework to identify genuinely tradable asset clusters. ORCA achieves superior performance across all key metrics. With an annualized return of 36.80%, it outperforms the strongest baseline, Contrastive Clustering (CC), by 1.6 percentage points. Crucially, this enhanced return comes with reduced risk—ORCA exhibits the lowest maximum drawdown at 17.18%, demonstrating the stability of our physics-informed clusters. Figure 3 illustrates the cumulative returns over the entire test period, showing ORCA’s consistent outperformance. The risk-adjusted metrics further validate our approach. ORCA attains the highest Sharpe

	AR	Vol	MDD	Sharpe	Sortino	Calmar
K-means	0.3054	0.1063	0.1904	2.5909	3.0118	1.4459
DBSCAN	0.2010	<u>0.1012</u>	0.1870	1.6893	2.0438	0.9144
Agglo	0.1462	0.0773	0.1948	1.5027	1.7852	0.5965
VAE	0.3200	0.1064	<u>0.1785</u>	2.7241	3.1496	1.6244
GAN	0.2530	0.1049	<u>0.2072</u>	2.1255	2.5197	1.0761
GAE	0.2321	0.1059	0.1892	1.9077	2.3583	1.0682
VaDE	0.2697	0.1047	0.2000	2.2901	2.6644	1.1982
CC	<u>0.3520</u>	0.1165	0.1815	<u>2.7640</u>	<u>3.1568</u>	<u>1.7741</u>
ORCA	0.3680	0.1176	0.1718	2.8747	3.3411	1.9669

Table 1: Performance comparison across clustering methods. ORCA consistently outperforms all baselines, achieving 36.80% annualized return with the lowest maximum drawdown (17.18%). Traditional clustering methods and market benchmarks show substantially inferior performance.

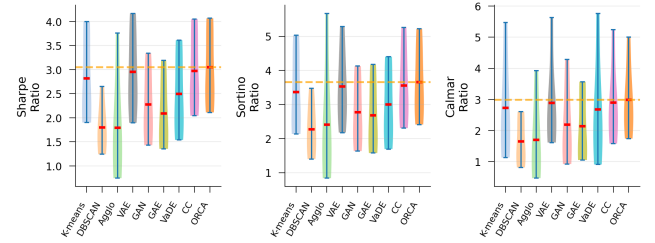


Figure 4: Temporal Stability of Risk-Adjusted Returns. Violin plots comparing the distribution of key performance metrics across non-overlapping 5-year periods.

ratio (2.87), Sortino ratio (3.34), and Calmar ratio (1.97), indicating that performance gains stem from improved cluster quality rather than increased risk exposure. The elevated Calmar and Sortino ratios specifically confirm that our OU process regularization effectively identifies clusters with robust mean-reverting properties. While traditional clustering methods exhibit low volatility, they fail to capture profitable trading opportunities, yielding inferior risk-adjusted returns. Even the state-of-the-art deep learning baseline (CC) falls short of ORCA’s performance across all metrics. These empirical results validate our core hypothesis: incorporating financial dynamics through physics-informed regularization produces asset groupings that are fundamentally more suitable for trading strategies.

4.3 Robustness Analysis

To assess the consistency of ORCA’s performance across varying market conditions, we partition the test period into non-overlapping 5-year intervals (2000-2004, 2005-2009, 2010-2014, 2015-2019, 2020-2023) and compute risk-adjusted metrics for each sub-period. Figure 4 presents the distribution of Sharpe, Sortino, and Calmar ratios across these intervals. ORCA demonstrates superior performance stability, maintaining the highest median values (indicated by horizontal bars) across all three metrics while exhibiting comparable variance to baseline methods. The compact distribution of ORCA’s performance metrics indicates robust returns without excessive

volatility across different market regimes, including the 2008 financial crisis and 2020 pandemic periods. Notably, while traditional clustering approaches show consistent but mediocre performance, and deep learning baselines (CC, VAE) exhibit higher variance, ORCA achieves an optimal balance between return magnitude and stability. This empirical evidence confirms that physics-informed regularization produces clusters with persistent mean-reverting properties, enabling consistent strategy performance independent of market conditions. The results validate that ORCA’s superior risk-adjusted returns are not artifacts of specific market environments but reflect fundamental improvements in cluster quality through the integration of financial dynamics into the learning objective.

4.4 Ablation Study

Model	AR	Vol	MDD	Sharpe	Sortino	Calmar
ORCA (Full Model)	0.368	0.118	0.172	2.874	3.341	1.966
– w/o $\mathcal{L}_{\text{PINN}}$	0.358	0.116	0.187	2.828	3.243	1.753
– w/o PLE	0.337	0.112	0.169	2.732	3.257	1.813
– w/o $\mathcal{L}_{\text{ins}}$	0.287	0.104	0.180	2.464	2.934	1.414

Table 2: Ablation Study of ORCA Components

We conduct a systematic ablation study to assess the contribution of each component in ORCA, with results summarized in Table 2. The analysis confirms that high-quality representation learning is foundational to our approach, as removing either the instance-level contrastive loss ($\mathcal{L}_{\text{ins}}$) or the Piece-wise Linear Encoding (PLE) leads to significant performance degradation. Most critically, the study highlights the unique role of the physics-informed regularization ($\mathcal{L}_{\text{PINN}}$). While its removal results in only a modest decline in the Sharpe ratio (2.873 to 2.828), it critically increases the Maximum Drawdown from 17.2% to 18.7%—an 8.7% relative increase in downside risk. This finding provides strong evidence that the primary function of the PINN regularizer is to enhance portfolio stability and control for downside risk by enforcing stable mean-reverting dynamics within the clusters. The complementary role of each component validates our integrated design.

4.5 Sensitivity Analysis

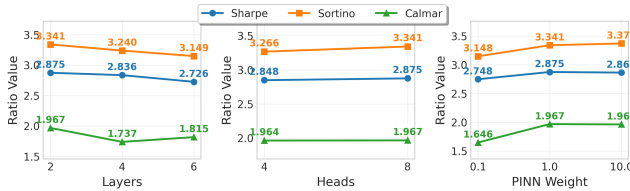


Figure 5: Sensitivity analysis of ORCA performance across key hyperparameters. Risk-adjusted metrics are evaluated while varying (a) the number of encoder layers, (b) the number of attention heads, and (c) the PINN weight β .

We examine ORCA’s sensitivity to key hyperparameters through systematic variation of architectural components and regularization

strength. Figure 5 presents the impact on risk-adjusted performance metrics. The model demonstrates remarkable stability across architectural variations. Varying the number of encoder layers from 2 to 6 yields only marginal changes in Sharpe ratio (2.875 to 2.726), with 2 layers surprisingly achieving the best performance. This suggests that the physics-informed regularization effectively guides learning even with minimal architectural depth. Similarly, increasing attention heads from 4 to 8 shows negligible impact on all metrics, indicating that the model does not require extensive attention mechanisms to capture relevant asset relationships. The PINN regularization weight β exhibits a clear optimal value at 1.0, where all three risk-adjusted metrics reach their peak. Underweighting ($\beta = 0.1$) reduces the Sharpe ratio to 2.748 and Calmar ratio to 1.646, confirming insufficient physics-based guidance. Conversely, overweighting ($\beta = 10.0$) slightly degrades performance, suggesting that excessive physics constraints can overshadow the data-driven learning. The convex relationship validates our choice of unit weighting as an effective balance between empirical patterns and theoretical priors.

5 Conclusion

This study introduces ORCA (Ornstein-Uhlenbeck Reversion and Contrastive Arbitrage), a novel approach designed to identify inherently tradable asset groups by integrating financial dynamics directly into the learning objective. ORCA leverages physics-informed regularization based on the Ornstein-Uhlenbeck process, thus ensuring that identified asset clusters exhibit genuine and stable mean-reverting properties conducive to statistical arbitrage. Empirical validation performed on U.S. equities from 2000 to 2023 demonstrates that ORCA consistently outperforms benchmark clustering methods, achieving notably superior risk-adjusted returns (Sharpe ratio: 2.87) and minimal maximum drawdown (17.18%), highlighting the value of incorporating domain-specific insights into representation learning.

The core innovation of ORCA resides in the concurrent optimization of a contrastive learning mechanism and a PINN regularizer within a single unified framework. Contrastive learning effectively captures complex nonlinear relationships among assets through deep representation learning, while the PINN module enforces adherence to stable mean-reverting dynamics. This integrative strategy directly addresses a principal challenge of traditional pairs trading methodologies—identifying asset clusters characterized by robust and actionable mean-reversion properties beyond mere statistical similarities.

Despite these advancements, several limitations warrant future investigation. Firstly, the fixed monthly rebalancing strategy potentially overlooks intramonth trading opportunities, suggesting future improvements could incorporate adaptive rebalancing schedules. Secondly, the static, threshold-based position selection mechanism currently employed is likely suboptimal under changing market conditions. Future research could beneficially explore reinforcement learning techniques to dynamically optimize position entry thresholds and portfolio allocations based on prevailing market dynamics and cluster stability metrics.

In conclusion, incorporating domain-specific physical constraints significantly enhances the effectiveness and practical applicability

of clustering-based trading methodologies. The encouraging results presented here highlight the broader potential of physics-informed machine learning methods in tackling complex quantitative finance challenges, providing a compelling direction for future interdisciplinary research.

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